

Probability Solutions - Q#02 and Q#03

Question 2

A survey shows that 70% of college seniors disapprove of daily pot smoking. Let X = number who disapprove among 12 seniors selected randomly. Then $X \sim \text{Binomial}(n = 12, p = 0.7)$.

(a) Probability that 7 to 9 disapprove

$$P(7 \leq X \leq 9) = P(X = 7) + P(X = 8) + P(X = 9)$$

$$P(X = k) = \binom{12}{k} (0.7)^k (0.3)^{12-k}$$

$$P(X = 7) = \binom{12}{7} (0.7)^7 (0.3)^5 \approx 0.254$$

$$P(X = 8) = \binom{12}{8} (0.7)^8 (0.3)^4 \approx 0.233$$

$$P(X = 9) = \binom{12}{9} (0.7)^9 (0.3)^3 \approx 0.164$$

$$P(7 \leq X \leq 9) \approx 0.254 + 0.233 + 0.164 = 0.651$$

(b) Probability at most 5 disapprove

$$P(X \leq 5) = \sum_{k=0}^5 \binom{12}{k} (0.7)^k (0.3)^{12-k} \approx 0.104$$

(c) Probability not less than 8 disapprove

$$P(X \geq 8) = \sum_{k=8}^{12} \binom{12}{k} (0.7)^k (0.3)^{12-k} \approx 0.485$$

Question 3

In a certain industrial facility, accidents occur infrequently. The probability of an accident on any given day is $p = 0.005$, and accidents are independent. Let X = number of days with an accident in 400 days. $X \sim \text{Binomial}(n = 400, p = 0.005)$.

Since n is large and p is small, we use **Poisson approximation**: $\lambda = n \cdot p = 400 \cdot 0.005 = 2$.

(a) Probability exactly 1 day with an accident

$$P(X = 1) = \frac{e^{-\lambda} \lambda^1}{1!} = e^{-2} \cdot 2 \approx 0.2707$$

(b) Probability at most 3 days with accidents

$$P(X \leq 3) = P(0) + P(1) + P(2) + P(3)$$

$$P(0) = e^{-2} \frac{2^0}{0!} = 0.1353$$

$$P(1) = e^{-2} \frac{2^1}{1!} = 0.2707$$

$$P(2) = e^{-2} \frac{2^2}{2!} = 0.2707$$

$$P(3) = e^{-2} \frac{2^3}{3!} = 0.1804$$

$$P(X \leq 3) \approx 0.8571$$

(c) Probability 10 to 15 days with accidents

$$P(10 \leq X \leq 15) = \sum_{k=10}^{15} \frac{e^{-2} 2^k}{k!}$$

Since $\lambda = 2$ is very small, these probabilities are extremely tiny:

$$P(10 \leq X \leq 15) \approx 0.000003$$